

Rings

Defn: A nonempty set R is said to be an associative ring if in R there are defined two operations denoted by $+$ and $\cdot$ respectively such that for all a, b, c in R :

1. $a+b$ is in R
2. $a+b = b+a$
3. $(a+b)+c = a+(b+c)$
4. There is an element 0 in R s.t. $a+0=a$ for every $a \in R$
5. There is an element $-a$ in R s.t. $a+(-a)=0$.
6. $a \cdot b$ is in R
7. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
8. $a \cdot (b+c) = a \cdot b + a \cdot c$ and $(b+c) \cdot a = b \cdot a + c \cdot a$.
(distributive laws).

Remark: 1. R is an abelian group under $+$.

2. A nonassociative ring is a ring in which associativity w.r.t. $\cdot$ fails.
3. A ring is commutative if elements commute with respect to the operation $a \cdot b$

We shall henceforth refer to $+$ as addition and $\cdot$ as multiplication.

Ex: 1. $R = \mathbb{Z}^{\leftarrow R, \mathbb{C}, \mathbb{Q}}$ with natural $+$ and $\times$.

2. $R = \mathbb{Z} \bmod 7$ under $+$ and $\times$ both mod 7.

$\hookrightarrow \{0, 1, 2, 3, 4, 5, 6\}$.

Obs: Non-zero elements form an Abelian group under $\times$.

Defn: If there is an elem 1 in R s.t for every a in R , $1 \cdot a = a \cdot 1 = a$, then such a ring is called a ring with unit element.

Obs: Ring may or may not contain 1 .

For example: Ring of even integers does not contain 1 .
(under usual $+$ and $\times$)

Ex: • (α, β)

$$x + y = (\alpha, \beta) + (\gamma, \delta) = (\alpha + \gamma, \beta + \delta)$$

$$x \cdot y = (\alpha, \beta) \cdot (\gamma, \delta) = (\alpha\gamma - \beta\delta, \alpha\delta + \beta\gamma)$$

• $\alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k$

real
Quaternions.

$$i^2 = j^2 = k^2 = ijk = -1$$

$$ij = -ji = k$$

$$jk = -kj = i$$

$$ki = -ik = j$$

$\mathbb{Q}$ is a noncommutative ring with $0=0$ and $1=1$.

Defn: 1. If R is a commutative ring, then $a \neq 0 \in R$ is said to be a zero-divisor if $\exists b \in R, b \neq 0$ s.t. $ab = 0$.

2. A commutative ring is an integral domain if it has no zero-divisors

3. A ring is said to be division ring if its nonzero elems form a group under multiplication.

Defn: A field is a commutative division ring.

Lemma: If R is a ring, then for all $a, b \in R$

1. $a \cdot 0 = 0a = 0$

2. $a(-b) = (-a)b = -(ab)$

3. $(-a)(-b) = ab$

If in addition, R has a unit element then

4. $(-1)a = -a$

5. $(-1)(-1) = 1$.

Lemma: A finite integral domain is a field.

Rings (Contd.)

- Defn: An element a in a ring is called unit if a^{-1} exists.

Note that we can do the following in rings unlike in groups.

$$xa = ya \equiv (x-y)a = 0 \rightarrow \text{It is not necessary that } x=y \text{ here.}$$

Theorem: If a ring has a unity, it is unique.

If a ring element has a mult. inverse, it is unique.

Pf: If there are two unities $1, 1'$ then

$$1 = 1 \cdot 1' \text{ and } 1 \cdot 1' = 1' \Rightarrow 1 = 1'.$$

If there are two inverses for an element a ,

say $ab = ba = 1$ and $ac = ca = 1$ then

$$b = b \cdot 1 = b \cdot ac = (ba)c = c \Rightarrow b = c.$$

Remark: We are not using $ab = ac$ ^{directly} to show that $b = c$.

Defn: An Integral domain is a commutative ring with unity and no zero divisors.

Defn: A field is a commutative ring with unity in which every non-zero elem is a unit.

Theorem: A finite integral domain is a field.

Pf: Guarantee: no zero divisors (from integral domain). Let the integral domain be D .

We need to show that

1. $1 \in D$ s.t. $a \cdot 1 = a \quad \forall a \in D$
2. For every elem $a \neq 0 \in D$ produce an elem. $b \in D$ s.t. $ab = 1$.

Let $x_1, x_2, \dots, x_n$ be elements of D . ~~suppose~~ Let a be a non-zero member of D . Consider $x_1 a, x_2 a, \dots, x_n a$. They are all in D (from closure under mult.).

We now claim that they are all distinct.

If $x_i a = x_j a$ then $(x_i - x_j)a = 0$. Since D has no zero divisors, it must be the case that $x_i = x_j$.

Thus all of $x_1 a, \dots, x_n a$ are distinct. Since a is an element of D , one of $x_1 a, \dots, x_n a$ must correspond to a . Let $\overset{x_i \text{ be s.t.}}{\uparrow} x_i a = a \Rightarrow (x_i - 1)a = 0 \Rightarrow x_i = 1$.

This gives us the identity in D . But we also need to show that x_i forms units w/ every element of D .

$$y \cdot x_i = (x_j a) x_i = x_j (\underbrace{a x_i}_{= x_i a}) = x_j a = y.$$

Say $y = x_j a$ for some j

Integral domain is a commutative ring.

Since $1 \in D$ then for every non-zero element we can find a mult. inverse through a similar procedure.

for a , x_k is a mult. inverse: $\underline{ax_k = x_k a = 1 = x_i}$

That is, there must be an elem corresponding to x_i in $\{x_1 a, \dots, x_n a\}$.

Repeat this argument for every non-zero elem a in D .

Other proof:

For an elem $a \neq 0$ in D , $a, a^2, \dots$ are in D as well.

From finiteness, we get that there must be

$$i, j \text{ s.t. } 0 \leq i < j \text{ s.t. } a^i = a^j \neq a^{j-i} = 1. \leftarrow$$

$$\downarrow$$
$$a^i = a^{j-i} \cdot a^i \neq a^i (a^{j-i} - 1) = 0$$

From this, a^{j-i-1} is inverse of a .

Corollary: $\mathbb{Z}_p$ is a field (p is prime).

Characteristic of a ring:

Defn: Let R be a ring with unity. If 1 has infinite order under addition then the characteristic is zero.

If 1 has order n under addition then the characteristic of R is n .

$$1 + 1 + 1 + \dots + 1$$

Thm: Characteristic of an integral domain is either 0 or prime.

Pf: That is, we need to show that the ^{additive order} char is either 0 or prime. For the sake of contradiction, assume that $n = st$ and 1 has order (additive) n .

$$0 = n \cdot 1 = (s \cdot t) \cdot 1 = (s \cdot 1)(t \cdot 1).$$

Either $s \neq 0$ or $t \neq 0$. We have no zero divisors.

Further n is the least +ve int with property $n \cdot 1 = 0$.

→ Either $s = n$ or $t = n$. Thus n is prime.

Ex: Roots of a polynomial $x^2 - 4x + 3 = 0$.

$$x^2 - 4x + 3 = (x-3)(x-1) = 0. \quad \left. \vphantom{x^2 - 4x + 3 = (x-3)(x-1) = 0} \right\} \text{No zero divisors}$$

Over $\mathbb{Z}$, we have two roots. Either $x=3$ or $x=1$.

On the other hand, over $\mathbb{Z}_{12}$ the no. of solutions is 4: $x=1, x=3, x=7$ and $x=9$. This is ~~bizarre~~ bizarre.

But with $\mathbb{Z}_{11}$ or $\mathbb{Z}_{13}$ things are fine.

Defn: Ideal:

A subring is called a (two-sided) ideal of R if for every $r \in R$ and every $a \in A$, both ra and ar are in A .

— Ideal absorbs the elements of R :

$$\forall r \in R: rA = \{ra \mid a \in A\} \subseteq A \text{ and } Ar = \{ar \mid a \in A\} \subseteq A.$$

Lemma: If ϕ is a homomorphism from R to R' with kernel K_ϕ then

1. K_ϕ is a subgroup of R under addition
2. If $a \in K_\phi$ and $r \in R$ then both ar and ra are in K_ϕ .

Pf: $K_\phi = \{a \in R \mid \phi(a) = 0_{R'}\}$. Point 1 follows from results in group theory.

2. $a \in K_\phi, r \in R$. Then $\phi(a) = 0$ and for ar , $\phi(ar) = \phi(a)\phi(r)$
 $= \phi(a) \cdot 0_{R'} = 0_{R'}$.
 $= \phi(a) \cdot 0_{R'} = 0_{R'}$.

Similarly for $\phi(ra)$.

$\Rightarrow K_\phi$ is an ideal.

Ex: 1. $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

2. If $\mathbb{R}[x]$ is set of all polynomials with real coeffs.

Let A denote the subset of all polynomials with const. term 0. Then A is an ideal of $\mathbb{R}[x]$ and $A = (x)$
or $\langle x \rangle$.

3. Let $a_1, a_2, \dots, a_n$ are elements of R (a comm. ring).

$I = \langle a_1, a_2, \dots, a_n \rangle = \{r_1 a_1 + r_2 a_2 + \dots + r_n a_n \mid r_i \in R\}$
Left ideal generated by $a_1, \dots, a_n$.

4. $\mathbb{Z}[x]$: Ring of all polynomials with $\mathbb{Z}$ coefs. --

I : Subset of $\mathbb{Z}[x]$ with even ~~coefs~~ ^{const.} only. $I = \langle x, 2 \rangle$.

Then I is an ideal of $\mathbb{Z}[x]$

Factor Rings:

$$R/A = \{r+A \mid r \in R\}.$$

Rings (contd.)

Subrings: A subset A of $\overset{\text{a ring}}{\uparrow} R$ that forms a ring under the same operations.

Ideal: A subring A of a ring R is called an ideal $\uparrow$ of R if for every $r \in R$, and every $a \in A$ both ra and ar are in A .

A is a proper ideal if A is a proper subset of R .

Obs: A is a normal subgroup of R under addition.

Factor rings: R be a ring, A be an ideal of R .

$$R/A = \{r+A \mid r \in R\}$$

Theorem: Let R be a ring and let A be a subring of R .

The cosets $\{r+A \mid r \in R\}$ is a ring under the operations

$(s+A) + (t+A) = s+t+A$ and $(s+A)(t+A) = st+A$ if and only if A is an ideal of R .

Proof: Note that a ring has closure under mult and is an Abelian group under addition.

It is easy to see that $\{r+A \mid r \in R\}$ forms a group under addition. We now need to show

1. If A is an ideal then mult. is well-defined.
2. If mult is well-defined then A is an ideal.

1. If A is an ideal:

$$\text{let } s+A = s'+A \text{ and } t+A = t'+A.$$

We need to show that $(s+A)(t+A) = st+A$ and $(s'+A)(t'+A) = s't'+A$ are Equal

$$\begin{aligned} st &= (s'+a)(t'+b) \\ &= s't' + at' + s'b + ab \end{aligned}$$

Since A is an ideal,
 $at' \in A, s'b \in A$

$$\Rightarrow st = s't' + c \quad \begin{matrix} st \\ \uparrow \\ c \in A \end{matrix} \quad c = at' + s'b + ab.$$

$$st+A = s't'+A.$$

Note that

$$s+A = s'+A$$

$$\Rightarrow s = s'+a \text{ for some } a \in A$$

$$t+A = t'+A$$

$$\Rightarrow t = t'+b \text{ for some } b \in A.$$

2. If A is a subring ^{of R} that is not an ideal.

Then $\exists a \in A$ and $r \in R$ s.t. $ar \notin A$ or $ra \notin A$.

Assume $ar \notin A$: since $a \in A, a+A = 0+A$.
 $r \notin A$.

$$(a+A)(r+A) = ar+A \text{ from the statement}$$

$$\text{but } (0+A)(r+A) = 0+A = A.$$

$\Rightarrow ar+A \neq A$ and thus our assumption that $ar \notin A$ is false.

Examples of factor rings:

$$1. \mathbb{Z}/4\mathbb{Z} = \{0+4\mathbb{Z}, 1+4\mathbb{Z}, 2+4\mathbb{Z}, 3+4\mathbb{Z}\}$$

$$(2+4\mathbb{Z}) + (3+4\mathbb{Z}) = 5+4\mathbb{Z} = 1+4\mathbb{Z}$$

$$(2+4\mathbb{Z})(3+4\mathbb{Z}) = 6 + \underbrace{20 \cdot 4\mathbb{Z}} + 4\mathbb{Z} \\ = 6 + 4\mathbb{Z} = 2+4\mathbb{Z}$$

$\mathbb{Z}/4\mathbb{Z}$ is arithmetic mod 4:

$$2. 2\mathbb{Z}/6\mathbb{Z} = \{0+6\mathbb{Z}, 2+6\mathbb{Z}, 4+6\mathbb{Z}\}$$

↑ Even numbers mod 6 arithmetic.

$$3. R = \left\{ \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \mid a_i \in \mathbb{Z} \right\}$$

I : Subset of matrices with even ^{$\mathbb{Z}$} entries.

→ Prove that I is an ideal.

$$R/I = \left\{ \begin{bmatrix} r_1 & r_2 \\ r_3 & r_4 \end{bmatrix} + I \mid r_i \in \{0,1\} \right\} \quad \text{Why?}$$

$$\underline{\text{Ex:}} \quad \begin{bmatrix} 5 & 3 \\ 2 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 2 & -4 \end{bmatrix} \\ \uparrow \in I$$

$$\neq \begin{bmatrix} 5 & 3 \\ 2 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + I \\ + I$$

4. $R = \mathbb{Z}[i] / \langle 2-i \rangle$

How does $\mathbb{Z}[i]$ look like?

$\hookrightarrow a+ib$

$\mathbb{Z}[i] / \langle 2-i \rangle = a+ib + \langle 2-i \rangle$

Recall that $\mathbb{Z}[i]$ denotes all polynomials over i with integer coeffs.

~~These get simplified with $i^2 = -1$.~~

$\sum_{k \geq 0} a_k i^k =$ Remainder upon dividing with $2-i$ + multiples of $(2-i)$.

Ex: $1+4i+3i^2$

$= 1+4i+3(i^2-4i+4) + 3 \cdot 4i - 12$

$= 1+16i+3(2-i)^2 + \cancel{3 \cdot 4i} - 12$

$= 1+16i-32+32-12+3(2-i)^2$

$= 21+16(i-2)+3(2-i)^2$

$= 21-16(2-i)+3(2-i)^2$

$= 21 + \langle 2-i \rangle$

5. $\mathbb{R}[x] / \langle x^2+1 \rangle$

$\langle x^2+1 \rangle = \{ f(x) \cdot (x^2+1) \mid f(x) \in \mathbb{R}[x] \}$

$\mathbb{R}[x] / \langle x^2+1 \rangle = \{ g(x) + \langle x^2+1 \rangle \mid g(x) \in \mathbb{R}[x] \}$
 $= \{ ax+b + \langle x^2+1 \rangle \mid a, b \in \mathbb{R} \}$

$\mathbb{R}[x] / \langle x^2+1 \rangle \cong \mathbb{C}$

[Cauchy]

Let R be a comm. ring w/ unity.

Then $\langle a \rangle = \{ ra \mid r \in R \}$

Principal Ideal generated by a .